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Title: An Electrochemical Synthesis Approach for Tetrasubstituted Olefins

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An Electrochemical Synthesis Approach for Tetrasubstituted Olefins

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Tianhao Wang, Yuying Zhang, Fangling Lu, Tao Wang, Zengzhan Yang, Yong Yuan (from left to right).

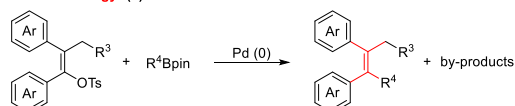
Mr. Fangling Lu is currently a first-year Ph. D. student within Prof. Ziwei Gao's research group at Shaanxi Normal University.

Mr. Yong Yuan is currently a third-year Ph. D. student within Prof. Aiwen Lei's research group at Wuhan University.

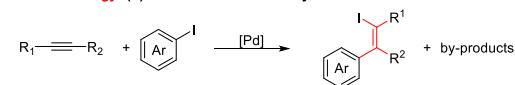
Since tetrasubstituted olefins serve as the key structure motifs in a large number of biologically active compounds, methods for synthesis of tetrasubstituted olefins are highly desirable. Among various synthetic methodologies, the most known strategies are the functionalization of olefins, alkynes, and allenes. For example, Zhang, Gosselin and co-workers prepared a series of tetrasubstituted olefins via Suzuki–Miyaura coupling (Scheme 1a).^[1] Morandi and co-workers reported a palladium-catalyzed intermolecular aryliodination of internal alkynes (Scheme 1b).^[2] Ma and co-workers prepared a series of tetrasubstituted olefins by using readily available organozinc reagents and 2,3-allenals (Scheme 1c).^[3] Despite extensive advances, the reported methods often require pre-prepared starting materials, inevitably leading to some by-products. Recently, under the supervision of Prof. Wang and Lei, we and Yang *et al.*, reported an unprecedented electrochemical oxidative Csp³-H/S-H cross-coupling reaction between thiophenols and acetonitrile (Scheme 1d), yielding a series of tetrasubstituted olefins under exogenous-oxidant-free conditions.^[4]

Scheme 1 Known knowledge and conceptual advance of this contribution.

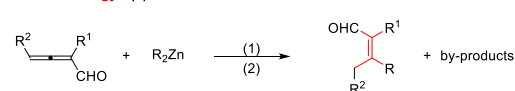
Known strategy: (a) functionalization of olefins



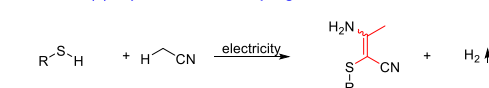
Known strategy: (b) functionalization of alkynes



Known strategy: (c) functionalization of allenes



This work: (d) Csp³-H/S-H cross-coupling



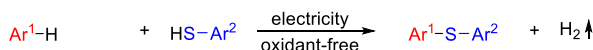
In 2017, our group reported an electrochemical oxidative C-H/S-H cross-coupling reaction between aryl/heteroaryl thiols and electronrich arenes.^[5] In 2019, our group demonstrated an

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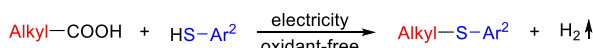
electrochemical oxidative C-H/S-H cross-coupling reaction of thiophenols with imidazopyridines.^[6] Under oxidant-free conditions, a series of diaryl thioethers were prepared in good to excellent yield (Scheme 2a). Inspired by these two works, we came up with the idea of developing an oxidant-free electrochemical method to prepare aryl alkyl thioethers (Scheme 2b).

Scheme 2 Electrochemical oxidative for C-S bonds formation.

(a) Electrochemical oxidative for the synthesis of diaryl thioethers

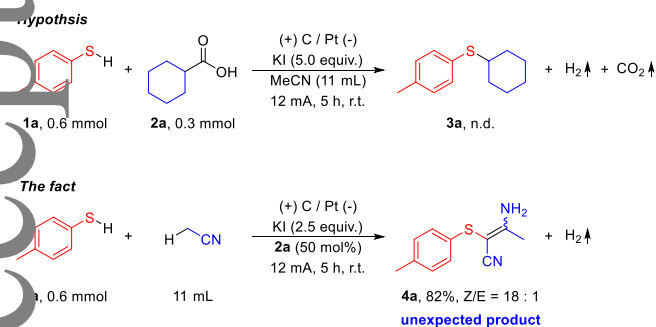


(b) Electrochemical oxidative for the synthesis of aryl alkyl thioethers



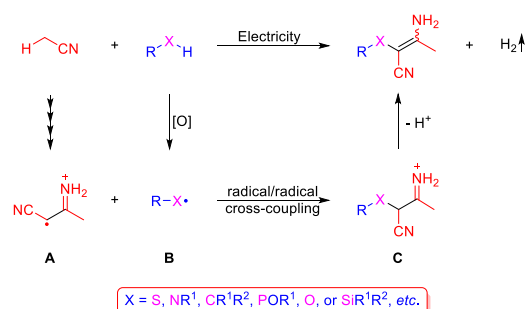
At the initial stage, 4-methylthiophenol and cyclohexanecarboxylic acid were employed as the model substrates to examine our concept. When the reaction was performed under a 12 mA constant current and using KI as the mediator, no expected C-S coupling product (**3a**) was obtained whereas an unprecedented Csp³-H/S-H cross-coupling reaction between thiophenols and acetonitrile was occurred and tetrasubstituted olefin product (**4a**) was obtained in 82% yield with excellent stereoselectivity (Z/E = 18 : 1) (Scheme 3). Encouraged by this interesting result, we turned our attention to the electrochemical oxidative Csp³-H/S-H cross-coupling reaction between thiophenols and acetonitrile to construct tetrasubstituted olefins.

Scheme 3 Electrochemical decarboxylation C-S bonds formation and electrochemical oxidative Csp³-H/S-H cross-coupling for synthesis of tetrasubstituted olefins.



During the study, the biggest challenge was the structural determination of unexpected product. After a long period of exploration, we obtained the single crystal of (Z)-3-amino-2-((4-fluorophenyl)thio)but-2-enitrile and finally determined the structure of the product by single crystal X-ray diffraction.^[7] During the preparation for publication of this research work, Huang and co-workers reported a similar results,^[8] which provided an extra-evidence for our observation.

Scheme 4 The strategy for synthesis various different tetrasubstituted olefins.



This reaction may provide a new strategy for synthesis of different tetrasubstituted olefins (Scheme 4). Under mild electrochemical conditions, acetonitrile is converted to the radical intermediate **A**, while thiophenols are oxidized to the corresponding radical intermediate **B**. Next, radical/radical cross-coupling produces the intermediate **C**, which undergoes further deprotonation leads to the desired product. Until now, we have only realized the synthesis of sulfur-containing β -enaminonitrile derivatives. We believe that by replacing thiophenols with other radical precursors, various different tetrasubstituted olefins, such as nitrogen-, carbon-, phosphorus-, oxygen-, or silicon-containing β -enaminonitrile derivatives, may also be obtained. Further application of this strategy for synthesis of tetrasubstituted olefins is under way in our laboratory.

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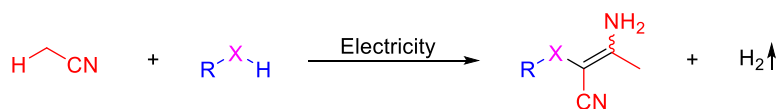
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We herein developed a new electrochemical synthesis approach for tetrasubstituted olefins.

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